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DYNAMIC MODELING OF TWIN-ROLL STEEL STRIP CASTING

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ABSTRACT

We consider the problem of dynamic coupling between the rapid thermal solidification and mechanical compression of steel in twin-roll steel strip casting. In traditional steel casting, molten steel is first solidified into thick slabs and then compressed via a series of rollers to create thin sheets of steel. In twin-roll casting, these two processes are combined, thereby making control of the overall system significantly more challenging. Therefore, a simple and accurate model that characterizes these coupled dynamics is needed for model-based control of the system. We model the solidification process with explicit consideration for the mushy (semi-solid) region of steel by using a lumped parameter moving boundary approach. The moving boundaries are also used to estimate the size and composition of the region of steel that must be compressed to maintain a uniform strip thickness. A novelty of the proposed model is the use of a stiffening spring to characterize the stiffness of the resultant strip as a function of the relative amount of mushy and solid steel inside the compression region. In turn this model is used to determine the force required to carry out the compression. Simulation results demonstrate key features of the overall model.

INTRODUCTION

Motivation and Problem Definition: Near-net-shape manufacturing processes are becoming a major contributor in the reduction of both environmental and economic costs in the industrial sector [1]. For the steel industry, twin-roll strip casting is one of the most prominent near-net-shape manufacturing processes.

It requires just one-tenth of the facility space, and it reduces the energy consumption by a factor of nine, as compared to traditional steel casting [2]. In the latter, molten steel is first solidified into thick slabs and then compressed via a series of rollers to create thin sheets of steel. In contrast, in twin-roll casting, molten steel is poured directly onto the surface of two casting rolls which simultaneously cool and compress the steel into a strip with a thickness of 1 – 3 millimeters. Combining these two steps into a single continuous casting process introduces coupling between the rapid thermal solidification dynamics and the mechanical stiffness of the resulting steel strip. To compensate for this coupling from a controls perspective, we require a simple and accurate model that characterizes the system dynamics.

Gaps in Literature: Many researchers have modeled the solidification process in twin-roll casting [3–7] but few have considered the coupling between the thermal and mechanical dynamics [5,6]. In order to design a controller that achieves the desired performance objective of uniform strip thickness, we require a simple model that captures the relevant input-output dynamics of the process. Santos et al. [3] and Liu et al. [4] created high-resolution simulations of the solidification process, but these are too complex to be used for control design. For example, Santos et. al derived a model with over 400 states. Furthermore, these models were intended to only capture the solidification process, and they do not examine the coupling between the solidification dynamics and the mechanical stiffness of the steel strip.

Other researchers have derived control-oriented models of the entire process [5–7]. However, their models assume an abrupt phase transition from liquid to solid steel when, in reality, this transition involves the storage of latent heat in a two-

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NOMENCLATURE

Symbol	Description	Symbol	Description
c	Specific heat	Ω	Rotational speed
f_m	Mushy fraction	Subscript	Description
F	Force	1	Liquidus
h	Heat transfer coefficient	2	Solidus
k	Thermal conductivity	<i>Comp</i>	Compression
L	Latent heat	g	Gap
R	Radius	k	Kiss
T	Temperature	ℓ	Liquid
x	Horizontal distance	<i>Lev</i>	Steel pool level
y	Height	m	Mushy
Z	Transverse direction	O	Outer boundary
ε	Strain	R	Steel at the roll
λ	Stiffness	<i>Roll</i>	Roll surface
ρ	Density	s	Solid
θ	Angle		

phase region known as mushy steel. The amount of mushy steel in the pool plays an essential role in determining the total force required to compress the steel into a uniform strip [8] and must be modeled to enable model-based control of the process.

Contribution: In this paper we describe a control-oriented dynamic model of the twin-roll steel casting process that accounts for multi-phase steel. A lumped parameter moving boundary approach is used to simplify the dynamics while still characterizing the composition of the steel in the nip region. A second model is derived to determine the force required to regulate the strip thickness.

Outline: The remainder of this article is organized as follows. We present a brief description of twin-roll strip casting in the Background section followed by an overview of the modeling approach for both the solidification and compression models. In the section titled Simulation Results, we demonstrate the features of each model. The paper ends with a discussion of our conclusions and future work.

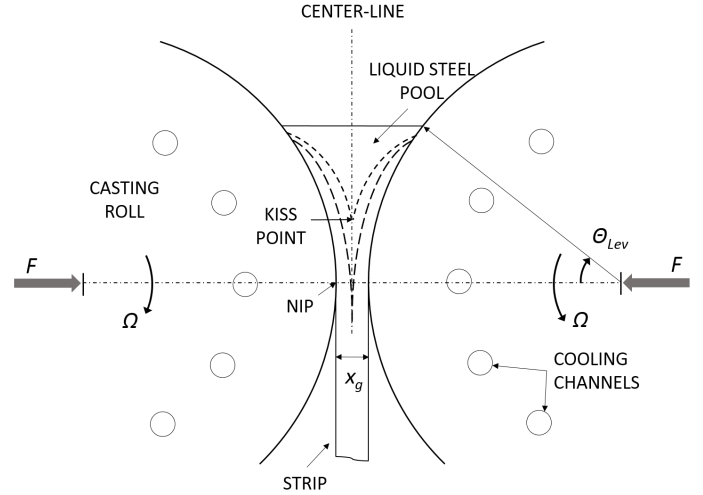


FIGURE 1. A TWIN-ROLL STRIP CASTER WITH DASHED LINES REPRESENTING THE BOUNDARIES BETWEEN THE LIQUID, MUSHY, AND SOLID STEEL PHASES.

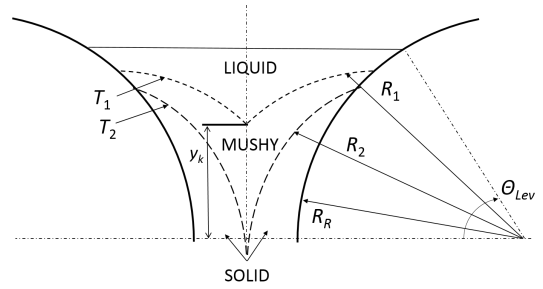


FIGURE 2. THE MOVING BOUNDARIES, R_1 AND R_2 , CORRESPONDING TO THE LIQUIDUS, T_1 , AND SOLIDUS, T_2 , ISOTHERMS RESPECTIVELY.

BACKGROUND

Figure 1 shows a schematic of the twin-roll casting process after it reaches steady state operation. During the casting process, molten steel is poured onto the surface of two casting rolls, thereby forming a liquid steel pool that covers the roll up to an angle θ_{Lev} . Energy is then transferred from the liquid steel, through the roll, and into a series of cooling channels. As this occurs, the liquid steel cools and begins to undergo a phase transition. Initially, it transitions to a semisolid, two-phase state known as mushy steel. As more heat is extracted from the pool, the mushy steel solidifies completely.

The phase transition occurs over the range of temperatures between the liquidus and solidus temperatures. All steel with a temperature above the liquidus temperature, T_1 , is considered to be completely liquid. Similarly, all steel below the solidus temperature, T_2 , is considered to be completely solid. *This means*

that the mushy region of the steel pool is defined as the volume of steel whose bulk temperature lies between the liquidus and solidus isotherms. With this distinction, the steel pool can be divided into three regions – liquid, mushy, and solid – with borders coinciding with the liquidus and solidus isotherms as shown in Fig. 2.

As each of the two rolls rotate, mushy and solid shells begin to form on their surfaces. These shells adhere to the roll surface and rotate with the roll while continuing to grow. When the micro-structures of the mushy and solid shells (from each roll) intersect at the center-line, they create a width of steel that may be wider than the desired strip thickness. When this occurs, the steel must be compressed to achieve the desired strip thickness. This compression, in turn, is achieved by applying a force, F , to both rolls in line with the *nip*, defined as the location at which the rolls are the closest to each other, as shown in Fig. 1. The amount of force required for the compression is influenced by the location of the intersection of the two mushy shells, called the *kiss point*. The distance from the kiss point to the nip is dependent on both the gap distance, x_g , and the rolls' rotational speed, Ω [5,9]. We then define the region of compression, also known as the *nip region*, as the volume between the nip and the kiss height, y_k . Within this region, the force required to compress the steel is dependent on the composition of steel which, in turn, is dependent on the growth rate of the solid steel shell and Ω .

From a control systems perspective, we are interested in modeling how the control inputs - x_g and Ω - affect our ability to manufacture a strip of steel of a desired thickness. To achieve this, we divide the process into two separate models. One model describes how the steel pool geometry and the rolls' rotational speed affect the composition of steel throughout the solidification process. That composition is then an input into a second model which calculates the force required to maintain a specified gap distance between the rolls.

SOLIDIFICATION MODEL

The solidification process is modeled using a lumped parameter moving boundary approach. The moving boundary approach is commonly used in models of multi-phase flows to track the boundary between different phases through the use of thermodynamic analysis [10, 11]. The twin-roll casting process is well suited to this approach because the steel pool can be divided into three distinct regions – liquid, mushy, and solid – with well-defined boundaries.

The governing equations for the solidification process are considered to be the same for each roll. As a result of this assumption, the steel pool is divided in half, and we model the process acting on one roll with the center-line of the pool treated as the outer boundary of each half pool. This geometry permits the half pool to be divided into smaller, discretized control volumes, which we will refer to throughout the manuscript as *slices*, with

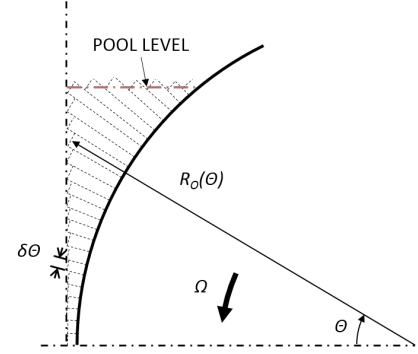


FIGURE 3. A SCHEMATIC OF A HALF STEEL POOL MODEL DIVIDED INTO CONTROL VOLUME SLICES WITH ANGULAR THICKNESS $\delta\theta$.

an angular thickness $\delta\theta$ as shown in Fig. 3. Within each slice, we assume that the mushy and solid shells *do not slip* along the roll's surface as the roll rotates. However, the liquid steel within each slice does not rotate with the roll; instead, the amount of liquid steel in each slice fluctuates to fill the entire volume.

Governing Equations

Within each slice we define a lumped parameter set that is used to calculate its dynamics through a combination of energy balances and boundary continuity equations.

Energy Balance. The energy balance for each region of steel is represented by the equation

$$\frac{\partial \rho c T}{\partial t} = \nabla (k \nabla T) + S, \quad (1)$$

where ρ is the density, k is the thermal conductivity, T is the temperature, and c is the specific heat corresponding to the steel phase. The variable S represents the heat source associated with phase transition and is only necessary in the mushy region. Within that region the heat source characterizes the release of latent heat needed to fully solidify the steel. Consistent with [3] and [12], the latent heat is characterized using the concept of pseudo-specific heat. However, here we take a simplified approach and assume that the amount of latent heat released during solidification is linearly related to the temperature of the mushy steel. The resulting pseudo-specific heat is given by

$$c_m = c + \alpha \frac{L}{T_1 - T_2}, \quad (2)$$

where T_1 is the liquidus temperature, T_2 is the solidus temperature, L is the latent heat of fusion, and α is a tunable parameter

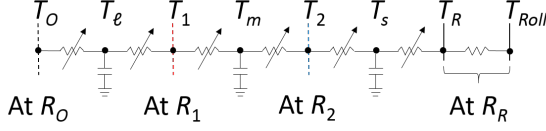


FIGURE 4. THE THERMAL IMPEDANCE MODEL FOR THE STEEL POOL.

that approximates the quality within the mushy region. For example, as α increases, the capacitance of the mushy region increases, signifying that there is a higher ratio of solid to liquid steel within the mushy region.

Within each slice, the total rate of change of the energy in each steel phase is obtained by integrating Eqn. (1) over the volume of each phase as shown in Eqn. (3). This equation is simplified by substituting $dV = r dr d\theta dz$ and applying the assumption that, for a given slice, the dynamics are uniform in both the angular (θ) and transverse (z) directions of the roll. The result is Eqn. (4).

$$\iiint_V \frac{\partial \rho c T}{\partial t} dV = \iiint_V \nabla \cdot (k \nabla T) dV \quad (3)$$

$$Z \delta \theta \int \frac{\partial \rho c T}{\partial t} r dr = \iiint_V \nabla \cdot (k \nabla T) dV \quad (4)$$

The left hand side of Eqn. (4) is simplified by applying Leibniz's rule and the product rule. The left hand side for the mushy region reduces to Eqn. (5). The liquid and solid regions are simplified similarly.

$$Z \delta \theta \left(\frac{\rho_m c_m}{2} (R_1^2 - R_2^2) \frac{dT_m}{dt} + \rho_m c_m (T_m - T_1) R_1 \frac{dR_1}{dt} \right) + Z \delta \theta \rho_m c_m (T_2 - T_m) R_2 \frac{dR_2}{dt} \quad (5)$$

The right hand side of Eqn. (4) is simplified by assuming that the heat transfer through the steel is dominated by the steel-roll interface and, thus, occurs only in the radial direction. Divergence theorem is then applied and the resulting simplification, in its most general form, is

$$\left(\frac{1}{r} \frac{\partial}{\partial r} \left(k Z \delta \theta r \frac{\partial T}{\partial r} \right) \right)_i + \left(\frac{1}{r} \frac{\partial}{\partial r} \left(k Z \delta \theta r \frac{\partial T}{\partial r} \right) \right)_o, \quad (6)$$

where k is the thermal conductivity of each phase of steel and the subscripts i and o denote the inner and outer radial boundaries of each phase.

Equation (6) is further simplified by assuming the process reaches quasi-steady state and by applying a thermal impedance

model similar to the one shown in Fig. 4. The resulting simplification is

$$\frac{k_p Z \delta \theta (T_i - T_p)}{\ln \left(\frac{R_p}{R_i} \right)} + \frac{k_p Z \delta \theta (T_o - T_p)}{\ln \left(\frac{R_o}{R_p} \right)}, \quad (7)$$

where the subscript $p \in \{\ell, m, s\}$ denotes parameters that are specific to the phase of steel being modeled.

Furthermore, the water passing through the roll is assumed to be an infinite heat sink, and the roll is considered to have no thermal capacitance. This results in a constant roll temperature that serves as an inner boundary for the steel pool model. This boundary is modeled by adding a contact resistance between T_R and T_{Roll} , as shown in Fig. 4.

Continuity Equation. The amount of energy crossing the liquidus isotherm should be the same for both the liquid and mushy regions. Likewise, the amount of energy crossing the solidus isotherm should be the same for both mushy and solid regions. We apply the following two continuity equations to ensure that these constraints are enforced.

$$0 = k_\ell \frac{T_\ell - T_1}{\ln \left(\frac{R_\ell}{R_1} \right)} - k_m \frac{T_1 - T_m}{\ln \left(\frac{R_1}{R_m} \right)} \quad (8)$$

$$0 = k_s \frac{T_2 - T_s}{\ln \left(\frac{R_1}{R_s} \right)} - k_m \frac{T_m - T_2}{\ln \left(\frac{R_m}{R_2} \right)} \quad (9)$$

In both Eqns. (8) and (9), R_ℓ is the radius corresponding to the center of the liquid phase and is calculated by $R_\ell = \sqrt{0.5(R_1^2 + R_O^2)}$. Similarly, R_m is the center of the mushy phase and R_s is the center of the solid phase, determined by $R_m = \sqrt{0.5(R_1^2 + R_2^2)}$ and $R_s = \sqrt{0.5(R_2^2 + R_R^2)}$, respectively. Equations (8) and (9) are true for all time. Thus, the time derivative of both equations is also equal to zero.

State Matrix Equation. The three energy balances and two time derivatives of the continuity equations combine to yield a system of state equations that describe the interactions between the state variables, R_1 , R_2 , T_s , T_m , and T_ℓ . The five state equations can then be written in the matrix form shown in Eqn. (10) with the non-zero elements listed in Table 1. This matrix equation applies for each slice, and the level of discretization can vary based on the needs of the engineer. As such, the total number of states needed to capture the solidification dynamics is five times the number of slices used.

TABLE 1. THE NONZERO VALUES OF THE STATE MATRIX EQUATION (10)

Element	Value
M_{12}	$\rho_s c (T_s - T_2) R_2$
M_{13}	$\frac{\rho_s c}{2} (R_2^2 - R_R^2)$
M_{21}	$\rho_m c_m (T_m - T_1) R_1$
M_{22}	$\rho_m c_m (T_2 - T_m) R_2$
M_{24}	$\frac{\rho_m c_m}{2} (R_1^2 - R_2^2)$
M_{31}	$\rho_\ell c (T_1 - T_\ell) R_1$
M_{35}	$\frac{\rho_\ell c}{2} (R_O^2 - R_1^2)$
M_{41}	$\left(-\frac{k_\ell (T_\ell - T_1) \left(\frac{0.5}{R_\ell} - \frac{R_\ell}{R_1^2} \right) R_1}{\ln \left(\frac{R_\ell}{R_1} \right)^2 R_\ell} + \frac{k_m (T_1 - T_m) \left(\frac{1}{R_m} - \frac{0.5 R_1^2}{R_m^3} \right) R_m}{\ln \left(\frac{R_1}{R_m} \right)^2 R_1} \right)$
M_{42}	$\frac{.5 k_m (T_1 - T_m) R_2}{\ln \left(\frac{R_1}{R_m} \right)^2 R_m^2}$
M_{44}	$\frac{k_m}{\ln \left(\frac{R_1}{R_m} \right)}$
M_{45}	$\frac{k_\ell}{\ln \left(\frac{R_\ell}{R_1} \right)}$
M_{51}	$\frac{0.5 k_m (T_m - T_2) R_1}{\ln \left(\frac{R_m}{R_2} \right)^2 R_m^2}$
M_{52}	$\left(-\frac{k_m (T_m - T_2) \left(\frac{0.5}{R_m} - \frac{R_m}{R_2^2} \right) R_2}{\ln \left(\frac{R_m}{R_2} \right)^2 R_m} + \frac{k_s (T_2 - T_s) \left(\frac{1}{R_s} - \frac{0.5 R_2^2}{R_s^3} \right) R_s}{\ln \left(\frac{R_s}{R_2} \right)^2 R_2} \right)$
M_{53}	$\frac{k_s}{\ln \left(\frac{R_s}{R_2} \right)}$
M_{54}	$\frac{k_m}{\ln \left(\frac{R_m}{R_2} \right)}$
A_1	$\frac{k_s (T_2 - T_s)}{\ln \left(\frac{R_s}{R_2} \right)} - \frac{T_s - T_{Roll}}{\ln(R_s/R_R) + \frac{1}{h R_R}}$
A_2	$\frac{k_m (T_1 - T_m)}{\ln \left(\frac{R_1}{R_m} \right)} - \frac{k_m (T_m - T_2)}{\ln \left(\frac{R_m}{R_2} \right)}$
A_3	$\frac{k_\ell (T_O - T_\ell)}{\ln \left(\frac{R_O}{R_\ell} \right)} - \frac{k_\ell (T_\ell - T_1)}{\ln \left(\frac{R_\ell}{R_1} \right)}$

$$\begin{bmatrix} 0 & M_{12} & M_{13} & 0 & 0 \\ M_{21} & M_{22} & 0 & M_{24} & 0 \\ M_{31} & 0 & 0 & 0 & M_{35} \\ M_{41} & M_{42} & 0 & M_{44} & M_{45} \\ M_{51} & M_{52} & M_{53} & M_{54} & 0 \end{bmatrix} \begin{bmatrix} \dot{R}_1 \\ \dot{R}_2 \\ \dot{T}_s \\ \dot{T}_m \\ \dot{T}_\ell \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ 0 \\ 0 \end{bmatrix} \quad (10)$$

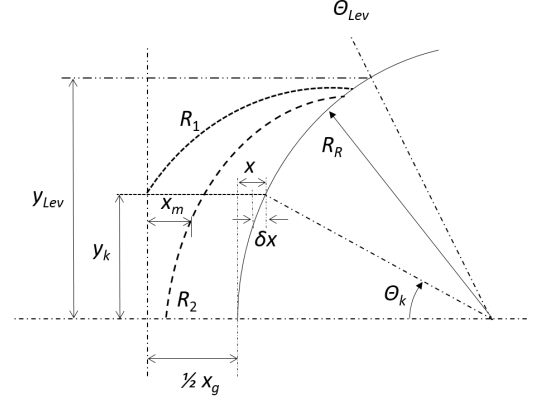


FIGURE 5. A SCHEMATIC OF THE MODEL OF THE COMPRESSION REGION.

COMPRESSION MODEL

After the steel solidifies into mushy and solid shells on the surface of the rolls, the compression model is used to characterize the force required to combine the two shells together within the nip region. The force interaction that we model is the force required to compress the shells in order to achieve a desired strip thickness. As the mushy and solid shells grow and rotate through the liquid steel pool, there is a point where the shells on both rolls have grown to the extent that they meet at the center-line of the pool and begin to weld together. When this occurs, as shown in Fig. 5, the combined shells create a strip that is $2x$ meters thicker than the gap distance, x_g . The steel must then be compressed to maintain the gap distance.

The force required to compress the steel is assumed to be highly dependent on the composition of the steel within the nip region. Therefore, we define a new parameter, the mushy fraction, f_m , such that

$$f_m = \begin{cases} 0, & \text{if } R_2 \geq R_O \\ \frac{R_O - R_2}{R_O - R_R}, & \text{if } R_R < R_2 < R_O \\ 1, & \text{otherwise} \end{cases} \quad (11)$$

Given the composition of the steel in the nip region, an estimate for the compressive force required by the process is determined using a mass-spring model of the steel strip. In this model we assume the stiffness of mushy steel is substantially lower than the stiffness of solid steel which means that the applied force will compress the mushy steel before compressing the solid steel. A stiffening spring model [13] is used to characterize this behavior. In the spring model, the force is a function of the stiffness of the steel, λ , and the compressive strain of the steel, ϵ_x . The compressive force, F_{Comp} , is calculated by integrating $dF_{Comp} = \lambda \epsilon_x dA$

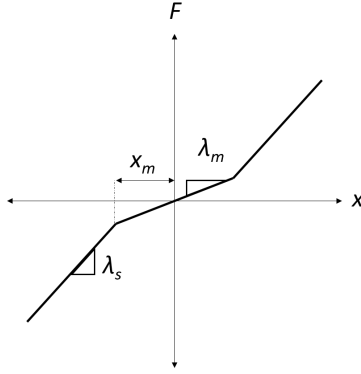


FIGURE 6. STIFFNESS CHARACTERISTICS OF THE STIFFENING SPRING AS IT RELATES TO THE STEEL COMPOSITION IN THE COMPRESSION REGION.

over the differential cross-sectional area of the compression region, dA . Substituting $dA = ZR_R \cos \theta d\theta$ and integrating both sides results in

$$F_{Comp} = Z \int_0^{\theta_k} \lambda(\theta) \varepsilon(\theta) R_R \cos(\theta) d\theta, \quad (12)$$

where the strain, ε_x is calculated as $\varepsilon_x = \frac{2\delta x}{2x+x_g}$ with δx and x defined as shown in Fig. 5.

The stiffness of the steel is dependent on the size of the mushy region, x_m , compared to the compressive displacement, δx . If δx is smaller than x_m , the compression is considered to only occur in the mushy steel as shown in Fig. 6. Once that threshold is exceeded, the stiffness increases because solid steel is being compressed in addition to mushy steel. The size of the mushy region is calculated using Eqn. (13) which relates the mushy fraction to the distance between the center-line and R_2 . The expression for the stiffness of the steel, λ , is then given by Eqn. (14).

$$x_m = (1 - f_m) \cdot \left(x + \frac{x_g}{2} \right) \quad (13)$$

$$\lambda = \begin{cases} \lambda_m, & \text{if } \delta x \leq x_m \\ \lambda_s, & \text{otherwise} \end{cases} \quad (14)$$

SIMULATION RESULTS

In this section we illustrate how the lumped parameter moving boundary approach captures the formation of both the mushy and solid steel shells with many fewer dynamic states than previous modeling efforts. Additionally, the results demonstrate how

TABLE 2. SIMULATION PARAMETERS

Parameter	Value	Units
R_R	250	mm
Inlet Steel Temperature	1600	C
T_{Roll}	150	C
x_g	2	mm
α	0.3	
Θ_{Lev}	48	deg
Ω	42.7	RPM

TABLE 3. STEEL PROPERTIES

Property	Value	Units	Property	Value	Units
c	750	J/kg C	T_1	1524	C
L	272000	J/kg	T_2	1502	C
k_ℓ	4000	W/m K	ρ_ℓ	7200	kg/m ³
k_m	4000	W/m K	ρ_m	7530	kg/m ³
k_s	77	W/m K	ρ_s	7860	kg/m ³

using the stiffening spring model characterizes the force required to compress the steel in the nip region as a function of the solidification dynamics.

For this case study we divide the steel pool into 6 slices which results in just 30 dynamic states. The model parameters used in the simulation are listed in Table 2, and the steel properties are listed in Table 3.

Solidification Results

In this case study, the liquid steel pool begins to cool and transition into mushy steel almost immediately after it comes into contact with the roll as shown in Fig. 7. The solid shell then gradually grows as more energy is extracted from the mushy steel. As a given slice of steel rotates toward the nip, the growth rate of both shells begins to slow. The slowing is explained by Fig. 8 which shows that the heat flux into the roll decreases as the slice rotates through the pool. The lower heat flux yields slower solidification and, thus, slower shell growth. The primary cause for the decreased heat flux is that the temperature of the steel contacting the roll, T_R , decreases over time. When the roll first rotates into the steel pool, liquid steel contacts the roll and T_R is equal to T_ℓ . Then, as the mushy and solid shells form, T_R begins tracking the temperature of whichever phase of steel is contacting the roll.

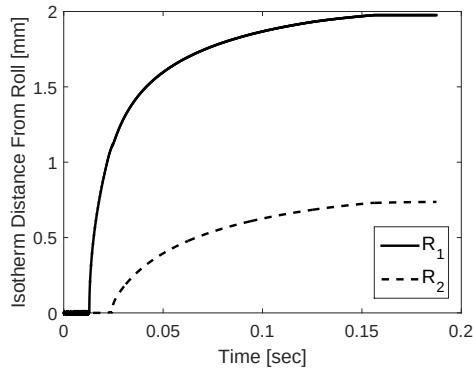


FIGURE 7. THE DISTANCE FROM THE LIQUIDUS (R_1) AND SOLIDUS (R_2) ISOTHERMS TO THE SURFACE OF THE ROLL FOR ONE SLICE AS IT ROTATES THROUGH THE STEEL POOL.

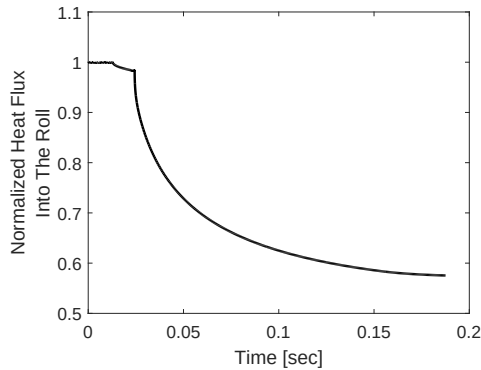


FIGURE 8. THE NORMALIZED HEAT FLUX EXTRACTED FROM ONE SLICE AS IT ROTATES THROUGH THE STEEL POOL.

The mushy and solid steel temperatures, T_m and T_s , are plotted as a function of time in Fig. 9. As expected, the mushy steel temperature varies little whereas the solid steel temperature decreases significantly as the roll rotates through the pool (approximately 0.19 seconds).

When we examine how all six slices evolve during the different stages of their rotation through the pool, we see the profile shown in Fig. 10. This profile shows that both the solid and mushy shells continue to grow as each slice gets closer to the nip, i.e. closer to $\theta = 0^\circ$. The mushy shell stops growing when it intersects the center-line, but the solid shell continues to grow. This behavior is similar to that described by Santos et al. [3].

This model also captures how changing α influences the growth of the mushy and solid shells. As α increases, the capacitance of the mushy region correspondingly increases. This affects the location of the liquidus and solidus isotherms as shown in Fig. 11. When α increases from 0.3 to 0.5 there is less solidification due to the increased capacitance, c_m . The higher c_m

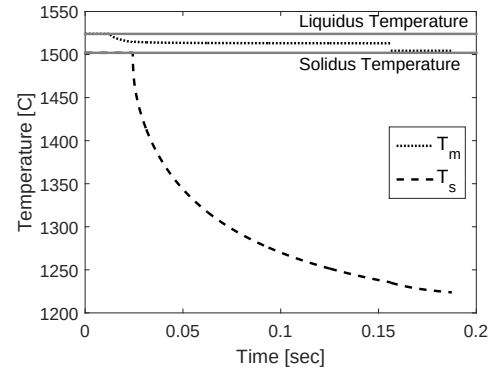


FIGURE 9. THE MUSHY AND SOLID STEEL TEMPERATURES IN ONE SLICE AS IT ROTATES THROUGH THE STEEL POOL.

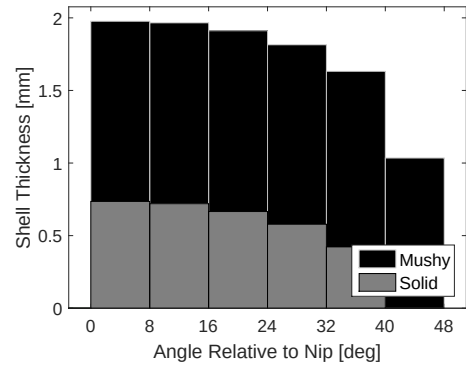


FIGURE 10. THE MUSHY AND SOLID SHELL PROFILES FOR ALL 8 SLICES AT THEIR RESPECTIVE LOCATIONS WITHIN THE STEEL POOL.

indicates that more energy must be extracted from the mushy region in order for the solid shell to grow. This capacitance varies with the quality of the mushy steel (i.e. the ratio of solid to liquid steel), but accurately modeling the quality would require a significantly more detailed model. Instead, α gives us a simple tuning parameter that we can use to approximate the steel quality and improve the accuracy of the model.

Compression Results

The nip region is formed when the mushy shell intersects the center-line of the pool as shown in Fig. 5. The mushy fraction, f_m , for the nip region is then calculated using the location of the solidus isotherm, R_2 , and Eqn. 11. With that information, the compressive force needed to maintain the gap distance, x_g , is determined by solving Eqn. 12.

The resulting force calculation is dependent on both the size of the nip region as well as f_m . As Fig. 12 shows, either an increase in the size of the nip region, calculated by θ_k , or a reduc-

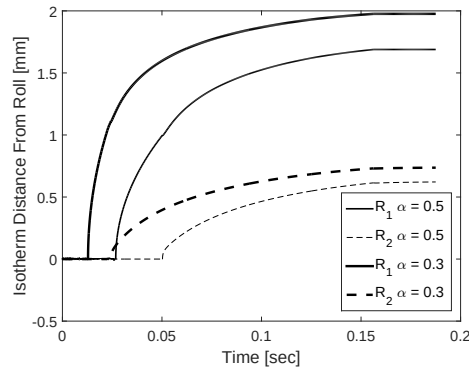


FIGURE 11. THE EFFECT OF VARYING α FROM 0.3 TO 0.5 ON THE LOCATION OF THE ISOTHERMS IN ONE SLICE AS IT ROTATES THROUGH THE STEEL POOL.

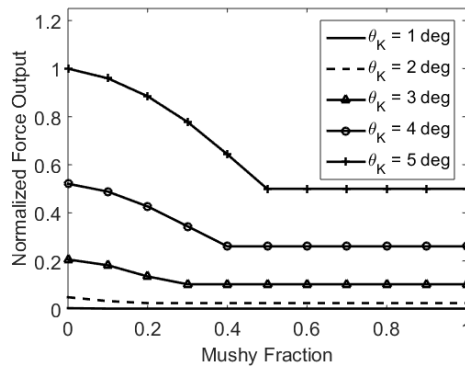


FIGURE 12. THE COMPRESSIVE FORCE IS A FUNCTION OF θ_{nip} AND f_m .

tion in f_m can result in an increase in the required force. Figure 12 also shows that for a very large f_m , the compression only occurs within the mushy steel. This results in a much lower force requirement than a nip region of the same size with a lower f_m (e.g. when the size of the solid shell is similar to the size of the mushy shell).

As the mushy and solid shell sizes change over time, both f_m and θ_k can fluctuate, causing the force requirement to also fluctuate. The proposed model gives us a method to calculate the change in the required force as a function of time.

CONCLUSION

This paper describes a control-oriented model of the twin-roll strip casting process. The proposed model effectively characterizes the three regions of the solidification process using only a fraction of the dynamic states needed in previous solidification models. The division of the pool into liquid, mushy, and solid

phases also assists in determining the size of the nip region and the amount of force required to compress the steel. By doing so, this modeling approach identifies the key coupling between the thermal solidification dynamics and the mechanical dynamics. In future work we will augment this model with the thermal and mechanical dynamics of the rolls into an overall model of the process that can be leveraged for multivariable control design. We will also investigate the effects that steel viscosity has on the compression model.

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